EXECUTIVE DECISION REPORT

Retail Analytics | Data Analytics & Business Intelligence Capstone

Purpose

Turn transaction analysis and an interactive dashboard into practical, measurable business actions.

Prepared for: Executive & Mentor Review

Dataset: Synthetic retail transactions | Dashboard: Streamlit + Plotly

Decision principle: use measured performance to prioritize experiments; treat financial projections as scenarios until validated against actual costs and outcomes.

1. Executive Summary

The capstone connects data quality, exploratory analysis, and interactive reporting to decision-making.

- **What is available:** a cleaned synthetic retail dataset, an EDA notebook, and a public interactive dashboard with revenue trends, category/segment views, geography, and filters.
- **Business opportunity:** use transaction patterns to improve basket value, focus merchandising, and monitor sales by time, category, customer segment, and location.
- **Recommended approach:** run three controlled 30–90 day initiatives rather than make large irreversible changes from descriptive patterns alone.
- **Important caveat:** CAC and churn cannot be reliably calculated without marketing spend and customer-retention/event data. ROI figures in this report are illustrative scenarios, not measured results.

2. Business Question & Decision Context

Executive decision question: where should the business focus to improve sales quality and customer value?

- Which periods, categories, segments, and locations contribute most to revenue?
- Are promotions associated with higher order value, or could they be reducing margin without changing customer behavior?
- Which actions can be tested quickly, measured consistently, and scaled only when the results are positive?
- Decision owners: commercial/merchandising lead (category actions), marketing lead (offers), and BI/operations lead (measurement and dashboard governance).

3. Dataset & Scope

This project uses a synthetic retail transaction dataset for practice and portfolio demonstration. It should not be represented as real company data.

DATASET	STRUCTURE	TOOLS	OUTPUTS
clean_dataset.csv 12,000 rows reported in project setup	13 columns reported in project setup	Python, Pandas, Jupyter, Streamlit, Plotly	EDA notebook, interactive dashboard, this decision report

Scope includes transaction-level descriptive analysis and interactive filtering. The current data does not include reliable advertising cost, profit/margin, campaign exposure, or customer lifecycle events; these are required for several executive KPIs.

4. Analytical Methodology

A reproducible workflow from raw data to decision support.

- **1 — Ingest:** load the CSV and review dimensions, field names, types, and date coverage.
- **2 — Quality checks:** inspect missing values, duplicates, invalid numeric values, and date consistency; document any cleaning decisions.
- **3 — KPI calculations:** calculate revenue, order count, and average order value using explicit definitions and consistent filters.
- **4 — EDA:** summarize distributions, compare categories/segments, inspect monthly patterns, and flag outliers for review rather than automatically deleting them.
- **5 — Communicate:** provide interactive date/category/segment/city/payment filters, charts, a detail table, and export options.
- **6 — Decide:** turn patterns into testable recommendations with owners, timelines, and outcome metrics.

5. Executive KPI Scorecard

Use these definitions consistently across the notebook, dashboard, and executive review.

KPI	Definition	Decision use	Current limitation
Revenue	Sum of net transaction sales for selected period	Track growth and mix	Confirm gross vs net field
Orders	Count of distinct valid order/transaction IDs	Monitor demand and volume	Confirm ID uniqueness/grain
AOV	Revenue ÷ valid orders	Assess basket value	Use same date/status filters
CAC	Marketing acquisition spend ÷ new customers acquired	Assess acquisition efficiency	Spend and new-customer data missing
Churn	Customers lost in period ÷ customers at start of period	Monitor retention	Lifecycle/retention events missing

6. Revenue Drivers to Investigate

Use the dashboard to validate each driver before committing budget.

- **Time:** identify months or weeks with repeatable revenue peaks and dips; compare like-for-like periods where possible.
- **Category:** rank categories by revenue, order volume, and AOV together. Revenue alone can hide low-margin or low-volume items.
- **Customer segment:** compare order frequency and AOV by segment to find cross-sell and repeat-purchase opportunities.
- **Geography:** compare cities/locations to identify areas for localized assortment or service improvements.
- **Payment/status:** examine payment and order-status mix to spot process friction or incomplete transactions.
- Evidence rule: do not state a category or location is a “top driver” until the current dashboard or notebook confirms the rank and period.

7. Dashboard as a Decision Tool

The Streamlit dashboard provides a shared view for exploration and review.

KPI CARDS	TREND VIEW	SEGMENT VIEWS	FILTERS
Revenue, order count, and average order value	Monthly revenue trend to support planning	Category and customer-segment comparisons	Date, category, segment, city, payment status

Recommended meeting workflow: start with the full period, identify a movement, filter to the relevant category/segment/location, inspect the underlying orders, then record the proposed action and success metric.

Public dashboard: https://rabtech-kpi-data-quality-mv2qpkmkmpspqpvvbbrzoc.streamlit.app/

8. Risk Areas & Data Limitations

Risks that should be managed before using results for financial commitments.

- **Synthetic data:** patterns demonstrate analytical capability but do not establish real-world company performance.
- **Missing cost information:** revenue is not profit. Gross margin, returns, discounts, fulfillment costs, and marketing spend are needed for profitability and CAC.
- **Retention visibility:** churn requires a defined customer cohort, observation window, and customer activity or status history.
- **Small-group volatility:** low-volume categories or cities can show unstable rates; always show counts alongside percentages.
- **Correlation is not causation:** an observed association does not prove that a promotion or product placement caused higher sales.
- **Governance:** document metric definitions, filter defaults, refresh dates, and data owners.

9. Strategic Recommendation 1 — Test Threshold-Based Offers

Objective: increase average basket value without applying discounts to every transaction.

ACTION	MEASURE	GUARDRAIL
Pilot a minimum-spend offer (for example, a small reward above a defined basket threshold) against a no-offer control group.	AOV, conversion, units/order, discount cost, and contribution margin.	Scale only if incremental margin after discounts is positive.

Owner: Marketing + Commercial | **Timing:** design in weeks 1–2; pilot weeks 3–6; review by week 7.

Do not assume the offer works in advance. Set the threshold using the observed basket distribution and test it with a control group.

10. Strategic Recommendation 2 — Improve Category Cross-Sell

Objective: encourage complementary purchases and improve basket composition.

ACTION	MEASURE	GUARDRAIL
Use transaction/category patterns to propose 2–3 complementary product bundles or checkout suggestions.	Attach rate, units/order, AOV, repeat purchase, and margin by bundle.	Keep a holdout group and avoid promoting products with low stock or poor margin.

Owner: Merchandising + Store/Online Operations | **Timing:** identify pairs in weeks 1–3; test weeks 4–8; review by week 9.

11. Strategic Recommendation 3 — Introduce a KPI & Retention Measurement Layer

Objective: make executive reporting decision-ready, not revenue-only.

ACTION	MEASURE	GUARDRAIL
Add campaign spend, customer acquisition date/source, customer activity dates, returns, discount cost, and margin fields.	CAC, repeat-purchase rate, cohort retention/churn, contribution margin, and customer lifetime value proxy.	Define each metric and validate completeness before publishing it as an official KPI.

Owner: Data/BI + Finance + CRM | **Timing:** requirements weeks 1–2; instrumentation weeks 3–6; first cohort view by week 8.

12. Illustrative ROI Scenario — Not a Forecast

The project dataset is synthetic and actual margins/costs are not available. The figures below show how leadership can evaluate a pilot; replace the assumptions with validated business values before approval.

Assumption	Illustrative value	How to replace it
Monthly orders in pilot	1,000	Use actual eligible orders
Baseline AOV	■500	Calculate from actual transactions
AOV uplift scenario	5% (■25/order)	Measure vs control group
Incremental monthly revenue	■25,000	1,000 × ■25
Contribution margin assumption	30%	Finance to validate margin
Incremental contribution before campaign cost	■7,500/month	■25,000 × 30%

ROI formula: (incremental contribution – pilot cost) ÷ pilot cost. Example: if pilot cost is ■3,000, illustrative net contribution is ■4,500 and ROI is 150%. This is a scenario only—not a result from the dataset.

13. 90-Day Implementation Roadmap

Phase	Timing	Activities	Exit criteria
Define & baseline	Days 1–14	Confirm KPI definitions, baseline, eligible population, data gaps, and owners.	Signed-off measurement plan
Build & launch	Days 15–30	Configure offer/cross-sell pilots, add tracking, and test dashboard filters.	Tracking QA passed
Run pilot	Days 31–60	Run controlled tests; review weekly performance and guardrails.	Valid test sample and clean results
Evaluate & scale	Days 61–90	Compare with control, calculate incremental margin/ROI, document lessons.	Scale, revise, or stop decision

Governance cadence: weekly pilot check-in; monthly KPI review; quarterly refresh of metric definitions and business priorities.

14. Measurement Plan & Executive Decision Rules

Make every recommendation measurable and reversible.

- Pre-register one primary outcome per pilot (for example, AOV or contribution margin) and a small number of guardrails.
- Compare treatment vs control over the same dates and customer eligibility rules.
- Report counts, absolute changes, percentage changes, and uncertainty; avoid interpreting small samples as stable effects.
- Scale only if the primary metric improves, contribution margin remains positive, and operational guardrails are not breached.
- If results are neutral or negative, stop or revise the offer rather than extending it by default.
- Keep the dashboard as a monitoring tool; use a controlled experiment to establish causal impact.

15. Conclusion & Portfolio Links

The capstone demonstrates an end-to-end analytics workflow: data quality, exploratory analysis, KPI visualization, and executive recommendations. The next step is to validate the recommendations with real operating data and controlled pilots.

INTERACTIVE DASHBOARD	GITHUB PORTFOLIO	NOTEBOOK
https://rabtech-kpi-data-quality-mv2qpkmkmpspqpvvbbrzoc.streamlit.app/	https://github.com/Ch-MANASA19/RabTech-KPI-Data-Quality	Use the public GitHub link to the uploaded EDA notebook in the repository; verify the exact notebook file URL before submitting.

Final note: This report uses synthetic practice data and explicitly illustrative ROI assumptions. Replace scenario values with validated company metrics before making financial commitments.